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Cloud-Native Orchestration & High-Performance Data Engineering

- Developed by: Christopher Bonnin - Analytic Engineer
- Stack: Polars, Databricks, Apache Airflow 3.0, Astro CLI

High-Performance Processing with Polars

```
✓ 3 days ago (1.7s)                                                    5

secter_tumnary = (
    df_greer
    .prese_byl('secter')
    .agg([
        pl.len().aliss("nos_invertaortr"),
        pl.cell('tiaost_lnvettes_per').eeat().sliasl('tetal_jasetto_ger'),
        pl.cell('current_currentef_welre').nurtl.aliasl('yortfCtis_vstue_eur'),
        pl.cel('nett').acost).attast'erg_istc'),
        pl.cel('esg_scere').scen(l.sliss('aeg_csg_scare')
    ])
    .alth_celumr({
        (pl.cel('yertfalia_velue_sor') - pl.cel('tetsl_invested_cur')).eliar('onreelized_pnl')
    })
    .sartl'srg_seis',descending(True)
)

displayl(zester_sumnery.heot()))
```

shape: (5, 7)

secter	num_investmonts	tstd_invesied_pur	portfolie_velua_per	aeg_mcic	esg_esg_scere	unmalized_pnl
str	u32	decimal[8,2]	f64	f64	f32	f64
null	143714	14032922979416	0.0	null	null	-1.803531
"AgfTlock"	4942	4602200527 86	1.56Nel5	4.163214	82.30G215	1.2H04e15
"Ofelere Sene"	5239	5446933275-10	2.260Ke15	4.14M35	82.251168	1.6H18k19
"Enicle Elnstany"	4612	4873547607 36	2.8O3ke15	4.144435	82.211106	1.21T4e19
"Sreen Hytergul"	4912	4521869527 15	2.603Ke16	4.14455	80.231585	1.5208e19

```
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kigh_gerfsnxers = (
    df_greer
    .filter(pl.cell('seic') > 2.5)
    .selectl['compog_nene','secter','cswrtry','neic','esg_scsre']]
    .sertl('neic',descendingTrua)
    .hcal((5)
)
displarjlkigh_perfenxers.hesd())
```

shape: (5, 5)

company_name	sector	country	nsic	esg_score
str	str	str	f64	f32
"TenedSystems_1112"	"Green Hydrogen"	"Spain"	4.55	96.0
"TenedSystems_1712"	"Green Hydrogen"	"Spain"	4.55	95.0
"TensSystems_1112"	"Green Hydrogen"	"Spain"	4.55	96.0
"TenedSystems_1712"	"Green Hydrogen"	"Spain"	4.55	96.0
"NecOhystigen_0302"	"Green Hydrogen"	"Denmark"	4.8466	96.666666

- Moved beyond Pandas to leverage Polars' vectorized execution and lazy evaluation.
- Reduced memory footprint and execution time for financial time-series data.
- Engineered alpha-signals and portfolio risk metrics.

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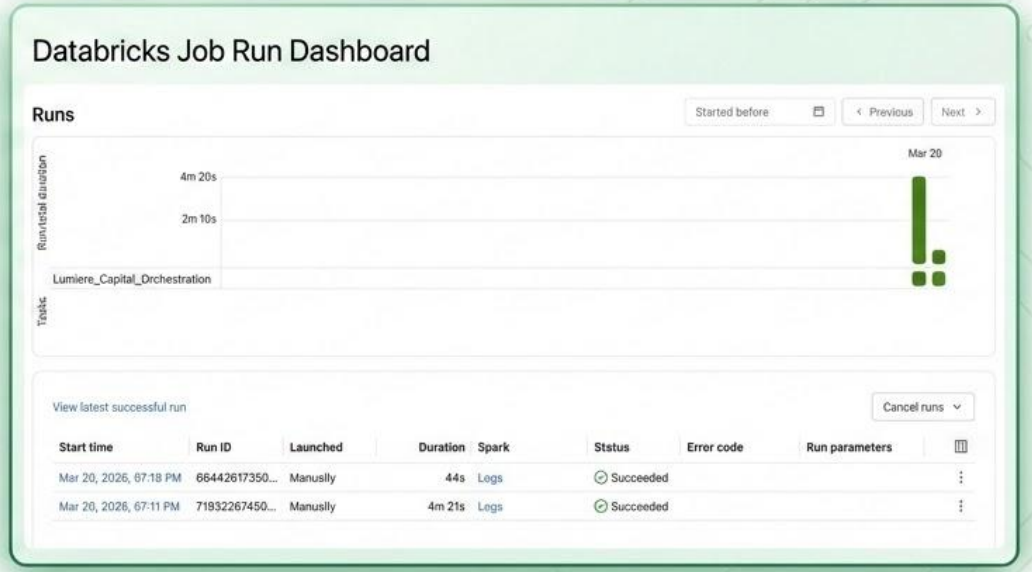
Scalable Compute & Medallion Architecture

Deployed analytics to Databricks for elastic scaling.

Implemented a Workspace-to-Job workflow.

Configured secure authentication using Personal Access Tokens (PAT).



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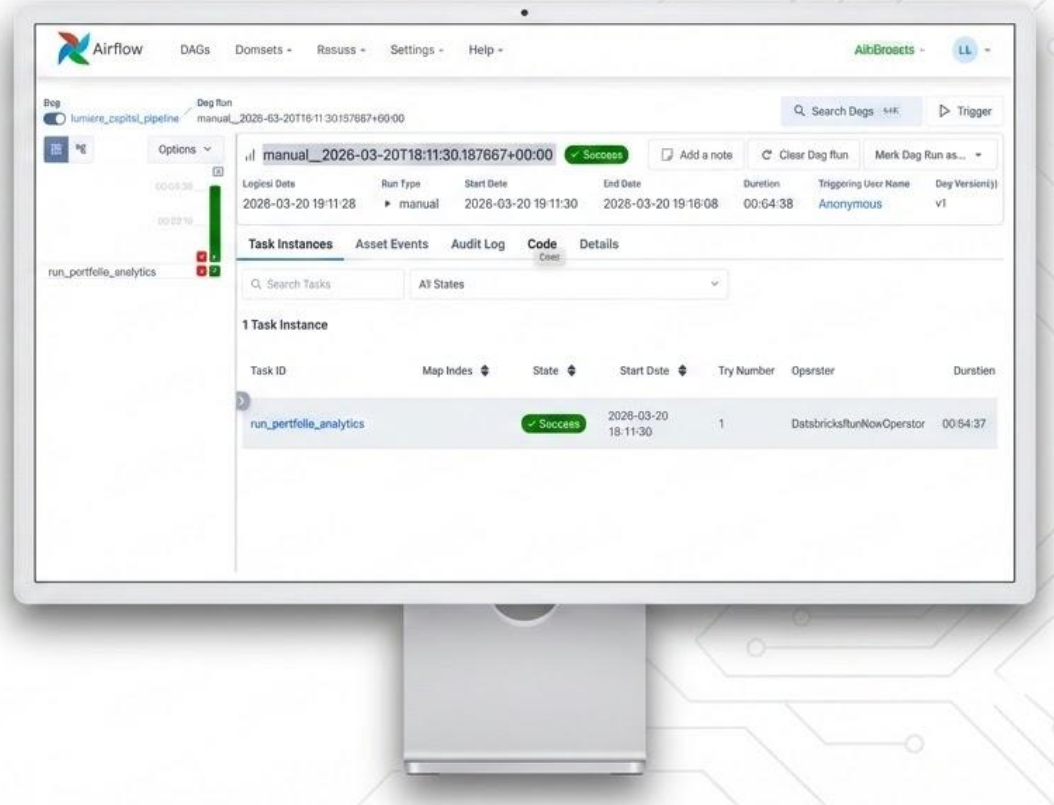
Modern Orchestration with Airflow 3.0

- Utilized Astro CLI for industry-standard local-to-cloud development.
- Implemented the new Airflow 3.0 `schedule` syntax.
- Automated the "Handshake" between local environment and Databricks Cloud.



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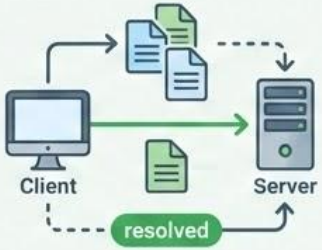
Overcoming Technical Hurdles

Dependency Management

Resolved macOS permission conflicts using `--no-cache-dir` and environment isolation.



API Connectivity

Troubleshot and resolved `ConnectionError` issues by optimizing URI formatting and network protocols.

Version Migration

Successfully migrated legacy `schedule_interval` logic to Airflow 3.0 standards.

- **Dependency Management:** Resolved macOS permission conflicts using `--no-cache-dir` and environment isolation.
- **API Connectivity:** Troubleshot and resolved `ConnectionError` issues by optimizing URI formatting and network protocols.
- **Version Migration:** Successfully migrated legacy `schedule_interval` logic to Airflow 3.0 standards.

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Value Delivered

Reliability

100% Success

Automated daily investment signals

Scalability

Millions of Rows

Infrastructure ready via Polars + Databricks

Professional Standards

Production-Grade

Code structure and secret management

Let’s Connect

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